

Representing Computable Biomedical Knowledge as FAIR Digital Objects to Promote Research and Clinical Decision Support

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COLLABORATIVE WORKSHOP

Abstract

Computable biomedical knowledge is knowledge related to human health and biology, represented in formats that can be processed or reasoned on by machines.[1] To advance data-driven clinical care and health research at scale, computable knowledge representations must be transparent, and reusable – this remains a significant challenge in a world where computable biomedical knowledge is proliferating alongside rapid advancements in AI.

To address this challenge, we have developed the Knowledge Object (KO), a metadata-driven digital object model for computable knowledge.[3,4] In current practice, computable knowledge implementations frequently rely on ad hoc solutions in which knowledge is tightly coupled to applications, often resulting in opaque systems that limit computable knowledge transparency and restrict its modification and reuse. In contrast, the KO model facilitates identification of and access to the knowledge, which in turn enhances the ability to manage, implement, modify, or reuse it, making it possible to apply existing KOs to new problems and contexts with or without changes.

This instructional workshop introduces the KO model and the practical benefits of this approach. Participants will explore a variety of use cases for computable biomedical knowledge, and the diverse ecosystem of stakeholders that support its development, management and application. Other topics include the role of standards-based metadata, and the practical implementation of the KO model in the form of larger Knowledge Assemblies, defined as purpose-built collections of KOs that fulfill specific tasks and which can enable efficient composition and reuse of computable knowledge across diverse contexts and use cases. Interactive workshop components will include a Google Colab Notebook tutorial, discussions, and a collaborative exercise to assemble a prototype Knowledge Assembly.

Topics to be covered

Foundations

1. Course and participant introductions – 10 minutes
2. Computable biomedical knowledge in clinical care, research, quality improvement – 15 minutes
3. Stakeholders in a computable knowledge ecosystem and their real-world needs – 15 minutes
4. Infrastructural approaches to describing and managing computable knowledge – 20 minutes
5. Relationships of FAIR Principles and Digital Objects to computable knowledge – 10 minutes
6. Standards-based metadata to describe computable knowledge – 20 minutes

Application and Practice

7. Use of Knowledge Objects: a model for FAIR representation of computable knowledge – 45 minutes
8. Use of Knowledge Assemblies: a model for FAIR and purposeful KO collections – 45 minutes

Interactive components

Hands-on experience: Course content relating to Knowledge Objects and Knowledge Assemblies is delivered via an easy-to-access Google Colab Notebook. The Colab Notebook demonstrates a computable knowledge use case in genomics with application to research, clinical practice, and knowledge management. Participants will use the Colab Notebook to a) load and run code to build and compute with Knowledge Objects, b) add metadata to existing Knowledge Objects, c) load and run code to build and compute with an operational Knowledge Assembly, d) use an LLM-enabled chatbot to find Knowledge Objects and interrogate their metadata.

Discussion: The course plan includes ample time for participant questions and discussion, both in small-group breakouts and large group sharing amongst the full workshop cohort. Discussion prompts will encourage participants to apply the course concepts and learnings to their own domains of expertise.

Group prototype exercise: Following completion of the Colab Notebook's well-organized modules, participants will work in small groups to collaboratively create a prototype Knowledge Assembly in a scenario-based exercise. This exercise will reinforce the core learning objectives about making computable knowledge more transparent and FAIR, while providing an opportunity for participants to contextualize the course concepts in diverse domains of expertise.

Educational objectives/outcomes

After participating in this session, attendees should be able to:

- describe the role of computable biomedical knowledge in a variety of domains, including: clinical practice, biomedical research, quality improvement, informatics implementation, knowledge management;
- identify and define diverse stakeholder roles in the biomedical knowledge ecosystem;
- summarize how metadata-driven models for computable biomedical knowledge can enhance FAIRness, transparency,

potential for reusability, and use at scale;

- relate approaches to making computable knowledge FAIR to real-world institutional needs;
- describe and discuss useful categories of metadata for computable biomedical knowledge;
- label and describe the components of a digital Knowledge Object;
- create selected metadata elements for a Knowledge Object;
- discover how rich metadata can enhance the findability of computable knowledge;
- employ LLMs and AI agents to answer questions about metadata-enriched computable knowledge;
- create a prototype of a purpose-built Knowledge Assembly from a variety of provided Knowledge Objects

Who should attend and prerequisites

This workshop is designed for anyone who creates, uses, implements, manages, or curates biomedical knowledge (including but not limited to guidelines, clinical decision support algorithms, clinical calculators, digital quality measures, computable phenotypes, etc.) including programmers, clinicians, health IT professionals, researchers, quality improvement professionals, and knowledge managers.

There are no prerequisites for this workshop. A laptop or tablet is strongly encouraged, and interaction with the Colab Notebook requires a Google account. Familiarity with running code in a Jupyter Notebook is helpful but is not required as an orientation to the Google Colab Notebook's functions will be provided.

Workshop instructional level

Introductory (70%), intermediate (30%)

Experience of instructors in targeted content areas

The workshop instructors are from the Knowledge Systems Lab (KSL), a health informatics lab in the University of Michigan Medical School's Department of Learning Health Sciences (DLHS).

Farid Seifi is a Senior Programmer in KSL, and earned his MS and PhD in computer science, with an emphasis on machine learning. Since joining KSL in 2023, he has focused his efforts on extending the Knowledge Object model, including creating KO-based tools, pipelines and chatbots, and modeling collections of KOs in preventive health services and pharmacogenomics. He co-created the Colab notebook that will be the basis for instruction.

Marisa Conte is a PhD candidate in Health Infrastructures and Learning Systems in DLHS, and the Associate Director of KSL. She has an MLIS with concentrations in health sciences librarianship and health IT. Marisa brings to KSL – and the workshop – nearly a decade of experience with data curation and the FAIR principles, open science practices, and metadata to enhance transparency and reuse, as well as a strong background in stakeholder engagement. Her research interests include open science, informatics implementation, and sociotechnical infrastructure for knowledge management and dissemination. Her PhD work examines computable phenotype transparency.

Allen Flynn is a Faculty Member at the University of Michigan Medical School and one of the founders of KSL. He has a strong

background in clinical decision support and computable knowledge organization and management. Dr. Flynn created the Knowledge Object Reference Ontology, from which KSL derived its current KO model. He has worked with KSL for 10 years to produce a wide variety of research outputs that collectively advance the formalization of Knowledge Objects and their metadata. He co-created KSL's initial instructional Google Colab notebook.

Previous presentations

An abbreviated (90-minute) online-only version of this workshop was presented and very positively received at the 2025 Global Meeting of the Mobilizing Computable Biomedical Knowledge (MCBK) community, and at the 2026 North America Chapter Meeting of this same community of interest.

References

1. Alper BS, Flynn A, Bray BE, Conte ML, Eldredge C, Gold S, et al. Categorizing metadata to help mobilize computable biomedical knowledge. *Learn Health Syst* 2022;6(n/a):e10271.
2. Knowledge Systems Lab, University of Michigan Medical School. Knowledge Object Implementation Ontology (KOIO) [Internet]. [cited 2025 May 8]; Available from: <https://biportal.bioontology.org/ontologies/KOIO>
3. Conte M, Flynn A, Barrison P, Boisvert P, Landis-Lewis Z, Friedman C. Digital objects to make computable biomedical knowledge FAIR: an infrastructural approach to knowledge representation, dissemination and implementation. *Res Ideas Outcomes* 2023;9:e109307.
4. Conte ML, Boisvert P, Barrison P, Seifi F, Landis-Lewis Z, Flynn A, et al. Ten simple rules to make computable knowledge shareable and reusable. *PLOS Comput Biol* 2024;20(6):e1012179.